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Dear Hiring Committee,

Thank you for considering me for the Senior Computational Biologist (Remote) position.

With the following, I aim to express my strong interest in the position within Research Computing at Boston Children's Hospital and highlight the most relevant skills and experience for the position.

I am currently a postdoctoral researcher at KU Leuven, where I develop computational methods for microbial network annotation [1] construct and curate databases of microbial growth phenotypes [2], and perform constraint-based metabolic modeling to investigate microbial interactions and functional capabilities [3–5].

During my Master's and PhD research, I designed and implemented HPC-oriented pipelines for amplicon [6, 7] and shotgun metagenomics [8] analyses, leveraging Singularity/Apptainer containers for reproducible execution on shared computing clusters. I also developed a microbial data integration platform that unifies diverse genomic and environmental datasets, enabling systematic analysis and interpretation of microbiome data across studies [9].

Since my PhD, I have been an active contributor to the open-source GeomScale project, where we develop advanced flux-sampling algorithms (in C++ and then wrapped in Python) for constraint-based metabolic models. As part of GeomScale's participation in Google Summer of Code, I have served as a mentor, guiding contributors in scientific software development and collaborative research coding practices (e.g. last year's project).

I have extensive experience with Git and GitHub, including managing collaborative repositories, reviewing contributions, coordinating distributed development, and maintaining reproducible, well-documented codebases.

In addition to research contributions, I have experience mentoring students and collaborators, preparing technical documentation, and delivering presentations tailored to audiences, ranging from computational experts to experimental scientists. I value clear communication, reproducible practices, and sustainable software design — principles that are essential for shared research computing services and long-term scientific impact.

I am excited by the opportunity to support cutting-edge biomedical research through robust data analysis pipelines, collaborative problem-solving, and scalable computing solutions.

I am particularly motivated by the mission of Research Computing at Boston Children's Hospital where my academic background can contribute directly to enabling scientific discoveries with tangible societal benefits.

Supporting investigators working with advanced technologies, such as single-cell genomics and spatial transcriptomics, represents a meaningful opportunity to apply my skills toward research with direct societal benefit, especially in the context of pediatric health.

I would welcome the opportunity to contribute my expertise in bioinformatics, scientific computing, and collaborative research to your team. Thank you for your time and consideration.

Haris Zafeiropoulos

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